Evidence-first automation for messy fraternity chapter data.

FratFinderAI crawls national fraternity directories, official school rosters, chapter websites, and social/contact sources, then verifies status and enriches records through queue-backed workflows with provenance and safety gates.

DATABASE MATURITY	OPERATOR SURFACE	LATEST LOCAL GATE
37 versioned Postgres migrations through operator audit events	**44** Next.js API route files protected by RBAC coverage tests	**589** crawler tests at 70.84% coverage, plus 47 web tests

How the system thinks

- **Discover** — national, school, chapter, search evidence
- **Extract** — tables, cards, scripts, locators, detail pages
- **Verify** — school-first status and identity checks
- **Enrich** — website, Instagram, email via safe queues
- **Explain** — provenance, conflicts, review, audit events

WHY THIS IS NOT A SCRAPER

> **The hard part is deciding which source should be trusted, what it proves, and whether the system is allowed to write.**

- **School recognition wins status.** National pages are useful but cannot override current official campus recognition.
- **Search is residual.** Search providers gather evidence; they are not the source of truth.
- **Ambiguity is a product state.** Weak evidence routes to review or blockers instead of unsafe canonical writes.

CORE STACK

- Next.js operator dashboard
- Python + LangGraph orchestration
- Postgres queues + provenance
- Status engine + school evidence
- SearXNG-first search reliability

What changed recently

STATUS	Campus recognition engine, conclusive absence, national capability profiles, and conflict-aware decisions.
QUEUE	Worker phase telemetry, typed blockers, cache-first school verification, and evidence refresh paths.
SEARCH	SearXNG endpoint failover, provider attempt history, degraded-mode protection, and smoke harnesses.
SECURITY	SBOM/SCA, SSRF-safe fetches, operator RBAC, and audit logging.

Validation snapshot

589 crawler tests · 47 web tests · 5 contract tests · 2 integration tests · Syft SBOMs · Grype SCA scan · CycloneDX artifacts · Route coverage

Security controls are documented in `docs/security/`, with detailed Phase 2 and Phase 3 validation reports plus a consolidated sprint report.

Safety gates that make the automation credible

Status-first contact writes

Inactive or unresolved chapters cannot receive newly discovered chapter contact fields by default.

SSRF-safe untrusted fetches

Blocks local/private/metadata targets, revalidates redirects, caps body size, and strips auth context.

Operator RBAC

Admin/operator/analyst roles gate mutating routes; privileged allowed, denied, and error outcomes are auditable.

Provider-aware degraded mode

Provider outages do not become false no-candidate outcomes; authoritative work can continue safely.

Provenance everywhere

Decisions and writes preserve source URLs, evidence type, confidence, reason codes, and review context.

Review over pollution

Ambiguous school, fraternity, chapter, or contact identity routes to review/blockers instead of silent writes.

BEST INTERVIEW FRAMING

FratFinderAI is an end-to-end data automation platform that handles unreliable public web sources using evidence ranking, queue orchestration, conservative write gates, and security controls. It is designed to be operated, audited, debugged, and improved from real benchmark results.

WHERE TO LOOK IN REPO

`docs/reports/PROJECT_REPORT.md` for the full project report. `docs/security/security-sprint-comprehensive-report.md` for DevSecOps proof. `CHANGELOG.md` for the implementation timeline.